

Scattered light may reveal the existence of ringed exoplanets

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ABSTRACT

Rings around giant exoplanets (hereafter ‘exorings’) are still a missing planetary phenomenon among the vast number of discovered planets. Despite the fact that there exist a large number of methods for identifying and characterizing these exorings, none of them has been successful to date. Most of those efforts focus on the photometric signatures produced by rings around transiting exoplanets; thus, little interest has been intended for the detectable signatures that non-transiting ringed planets might cause owing to the excess of scattered starlight from both their atmosphere and the considerably large surface of their (hypothetical) ring system. This extra scattering produced by exorings would occur at an orbital location defined here as ‘the summer solstice’ of a stellar light curve. In this letter, we develop a first-order model to estimate the photometric signatures of non-transiting exorings, and predict their detectability by using present and future facilities. We also show how, besides the discovery itself, our model can be used to constrain orbital and physical parameters of planet–ring systems.

Key words: methods: analytical – techniques: photometric – planets and satellites: rings.

1 INTRODUCTION

Thousands of exoplanets discovered so far are constantly unveiling new planetary phenomena and moving farther the limits of our knowledge. In not more than 35 yr, humanity has discovered and characterized planets with masses and sizes spanning from a tenth of Mercury to tens of Jupiter, and within a wide range of orbital configurations, compositions, and dynamical environments (see Perryman 2018). All of these discoveries have been accomplished with the combined power of ground- and space-based telescopes and spectrographs (e.g. *Kepler*, *TESS*, *SuperWASP*, *HARPS*, etc); besides, the advent of valuable data from exoplanet projects/instruments such as *PLATO*, *CHEOPS*, *SPHERE/ZIMPOL*, *E-ELT*, *GMT*, and *JWST* (see e.g. Lopez-Morales et al. 2019) portends significant upcoming achievements.

In the last decade, the systematic monitoring of stellar light curves has proven high success for the detection and characterization of exoplanets. This technique, known as transit photometry, has enabled the study of planetary atmospheres (see e.g. Batalha, Kempton & Mbarek 2017; Mayorga et al. 2019), detected exocomets and unusual objects (see e.g. Rappaport et al. 2018), suggested the

existence of yet-to-be-confirmed exomoons (Teachey & Kipping 2018), found debris discs around substellar objects (Kenworthy & Mamajek 2015), and may have the potential of revealing other previously unknown planetary processes as those in Boyajian et al. (2016) and Sucerquia et al. (2019). Other methods like direct imaging and submillimetre interferometry have also confirmed their huge potential. In recent years, the detection of complex circumstellar disks that hint at unusual planetary formation processes, as well as *circumplanetary discs* where moons or even planetary rings may possibly form (see e.g. Isella et al. 2019), has been reported.

Contrary to the outer planets of our Solar system, the detection of rings around giant exoplanets (‘exorings’ for short) remains elusive. These exorings may produce notorious photometric signatures if they orbit around transiting exoplanets (see Zuluaga et al. 2015 and references therein), but no effort has fully succeeded at detecting the first exoring (see e.g. Piro & Vissapragada 2019). However, transit photometric signatures are not the only viable method for identifying exorings, as it will be demonstrated later in this work.

In addition, by studying highly polarized scattered light from planetary surfaces, projects like *RefPlanets* (Hunziker et al. 2020) intend to perform the search, discovery, and characterization of evolved cold planets around nearby (bright) stars, making use of the instrument *SPHERE/ZIMPOL* attached to the VLT (Schmid et al. 2018). Scattered light has also been used for studying the transiting

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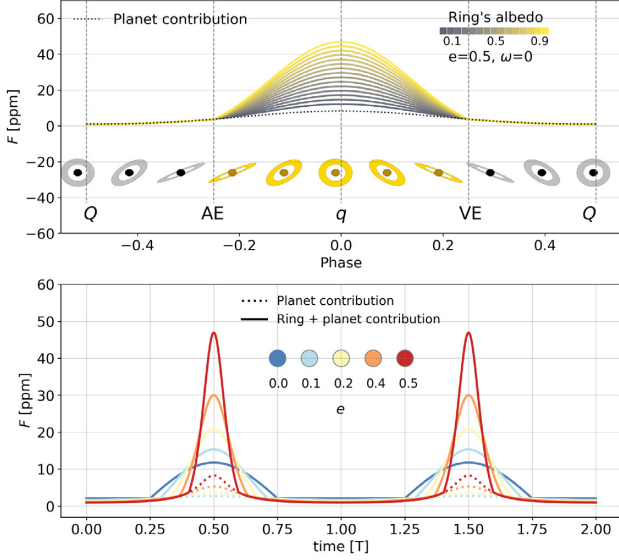


Figure 2. Upper panel: Normalized stellar flux of an exoring during one planetary orbit for albedos ranging from 0.1 to 1. Icons below the curve represent the cross-section of the ring system as seen from the star, where their colours indicate if the side towards the observer is illuminated (yellow) or not (grey). Lower panel: Bright curve for two orbital periods showing the contribution of the planet (dotted lines) and planet + ring (solid lines).

au from a Sun-like star, and assuming $\alpha_{rp} = 1$ and $i = 39^\circ$. It is worth mentioning that the actual axis tilt of Saturn's rings is $\sim 27^\circ$, and $i \gtrsim 39.2^\circ$ leads to Lidov–Kozai oscillations caused by stellar gravitational perturbations (Sucerquia et al. 2017). The stellar light curve in the lower panel of Fig. 2 allows us to identify four important properties: (1) the *maximum flux anomaly*, $F_{\max}$, or height of the peaks; (2) the *bright interval*, t_{bright} , or width of the peaks, corresponding to the time spent by the planet at VE and AE (when the illuminated side of the ring can be perceived by the observer); (3) the *dark interval*, t_{dark} , or time between AE and VE; and (4) the *dark flux anomaly*, F_{dark} , defined as either the maximum or minimum value of F during t_{dark} (see Section 3).

From the solid and dashed curves in the lower panel of Fig. 2, it can be noticed that the maximum flux anomaly of the ring is almost one order of magnitude larger than that of the planet, at least for the conditions considered in our toy-model (i.e. face-on orbit and summer solstice happening at periastron). This implies that for some current observing campaigns, the presence of exorings could be detected when looking for scattered light, and at the same time, the absence of this conspicuous signal may allow to constrain their occurrence rate among discovered extrasolar systems.

Contour plots of the maximum flux anomalies $F_{\max}$ are shown in Fig. 3 for a system with similar properties but different orbital configurations from those of Fig. 2. Within the conditions considered in the toy model presented here, Fig. 3 indicates that only those rings around close-in giant planets might be detected by *TESS*, *PLATO*, *CHEOPS*, and *JWST*. However, if future instruments achieve photometric sensitivities well below the ~ 10 ppm level, cold planets (even beyond the snow line) are theoretically detectable via scattered light from their system of rings.

3 FROM BRIGHT CURVE TO ORBIT

Can orbital and physical properties of non-transiting ringed planets be determined, estimated, or constrained from their light-curve

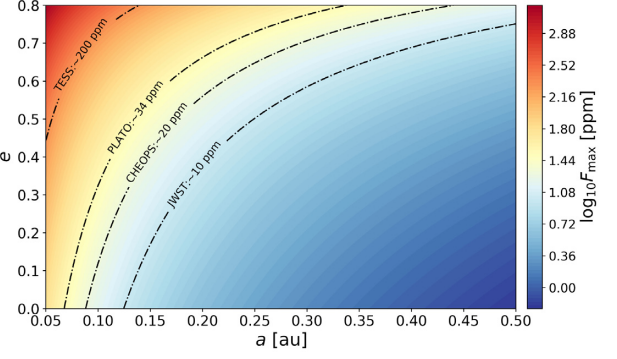


Figure 3. Contour plot of the normalized stellar flux anomaly F in the orbital parameter space (e – a). We assume a Sun-like star and Kronian properties for the planet and its rings.

attributes? Light curves certainly depend on many parameters in addition to those considered in our model; thus, the solution to this problem will be highly *degenerated*. However, we can still constrain the basic features of the system through the four main properties defined in the previous section. At the research stage of this topic, where only the simplistic toy model of Section 2 has been developed, we will follow a parallel line of thought to the seminal work on transiting exoplanets by Seager & Mallén-Ornelas (2003).

We assume that the mass of the star, M_* , is much larger than that of the planet, M_p (i.e. $M_* \gg M_p$), and that M_* can be estimated from independent methods such as the mass–luminosity relationship. Still, estimating accurate (higher than the mass–luminosity relationship) masses and radii for main-sequence stars from observations is not a straightforward task. It usually involves a combination of asteroseismology and/or interferometry, with observables calibrated against eclipsing binaries (see e.g. Moya et al. 2018). However, space-based missions are providing a very rich wealth of data that allow (or will allow shortly) for these precise estimates to be calculated in an almost routine fashion.

The planet's orbital period, T , will be equivalent to the interval between light-curve peaks (see Fig. 2), so the semimajor axis is computed as $a \approx [GM_*(T/2\pi)^2]^{1/3}$ (i.e. Kepler's third law). In our toy model and under the assumption of a pure Keplerian undisturbed orbit, the width of the bright peaks in the light curve is the time from VE ($f = -\pi/2$) to AE ($f = \pi/2$) and is simply given by

$$\frac{t_{\text{bright}}}{T} = \frac{1}{2\pi} \left[\cos^{-1} \eta_+ + \cos^{-1} \eta_- - e \left(\sqrt{1 - \eta_+^2} + \sqrt{1 - \eta_-^2} \right) \right], \quad (5)$$

where

$$\eta_{\pm} = \frac{e \pm \sin \omega}{1 \pm e \sin \omega}, \quad (6)$$

with ω the periastron angle. If $\omega = 0^\circ$,

$$\frac{t_{\text{bright}}}{T} = \frac{1}{\pi} \left[\cos^{-1}(e) - e\sqrt{1 - e^2} \right], \quad (7)$$

which can be numerically inverted to provide e as a function of t_{bright} . For small values of e , we can write an analytical expression for e as a function of t_{bright} (see the dashed line in Fig. 4):

$$e \approx \frac{\pi}{2} \left(\frac{1}{2} - \frac{t_{\text{bright}}}{T} \right). \quad (8)$$

This linear approximation always underestimates the actual value of e , and up to $e \approx 0.5$ it has relative errors below 5 per cent.

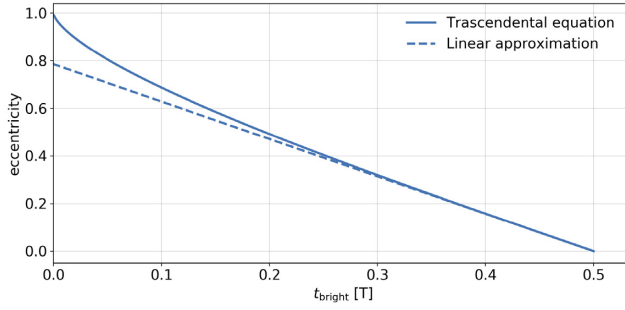


Figure 4. Planetary eccentricity (e) as a function of the bright time (t_{bright}) in the toy model developed in this work (see equation 7).

To estimate the planet's size (or more precisely its Lambertian area), we can use the dark flux anomaly F_{dark} . From equation (1), between AE and VE, the flux $F(t)$ is constrained by

$$F(t) \leq F_{\text{dark}} \equiv \frac{\alpha_p A_{\text{planet}}}{4\pi a^2 (1 - e^2)^2}. \quad (9)$$

Since $\alpha_p \leq 1$ we can constrain the planetary Lambertian area:

$$\frac{A_{\text{planet}}}{a^2} \geq \frac{A_{\text{planet, min}}}{a^2} \equiv 4\pi (1 - e^2)^2 F_{\text{dark}}. \quad (10)$$

In our toy model (see Section 2), F_{max} occurs at periastron $q = a(1 - e)$. Thus, according to equation (1) F_{max} is given by

$$F_{\text{max}} = \frac{\Delta_{\text{pr}}}{4\pi (1 - e)^2}, \quad (11)$$

where

$$\Delta_{\text{pr}} = \alpha_r \frac{A_{\text{ring}}}{a^2} \cos i \sin i + \alpha_p \frac{A_{\text{planet}}}{a^2}. \quad (12)$$

As $\alpha_r, \alpha_p \leq 1$, $\cos i \sin i < 1$, and $A_{\text{planet}} < A_{\text{planet, min}}$, we have

$$\frac{A_{\text{ring}}}{a^2} \geq \frac{A_{\text{ring, min}}}{a^2} \equiv 4\pi (1 - e^2) F_{\text{max}} - \frac{A_{\text{planet, min}}}{a^2} \quad (13)$$

To sum up, if we measure F_{max} , F_{dark} , and t_{bright} from the light curve of a non-transiting ringed exoplanet, we can compute e (equation 8), $A_{\text{planet, min}}/a^2$ (equation 10), and $A_{\text{ring, min}}/a^2$ (equation 13). Additionally, if we find a from T and M_* , an estimation of the planet and rings' radii can also be obtained.

4 BEYOND THE TOY MODEL

The toy model developed so far has assumed a very specific orientation for the orbit of a non-transiting planet and its rings. This simplifies significantly the analytical description of its light curve in a similar avenue to the original modelling of transit light curves by Seager & Mallén-Ornelas (2003). Two basic quantities, however, will certainly be different in real systems: (1) ω with respect to the summer solstice (i.e. point q in Figs 1 and 2) and (2) a non-zero inclination i of the orbit respect to the sky plane.

The effect that ω has on the light curve is shown in the upper panel of Fig. 5. It is evident that as the summer solstice shifts its position from the periastron, the amplitude of the signal decreases and reaches a minimum when the summer solstice occurs at $\omega = 180^\circ$ (i.e. the apoastron). Still, the light curve preserves its main features having a fainter F_{max} , a larger F_{dark} , and a t_{bright} much harder to measure, as compared to the toy model.

In the lower panel of Fig. 5, we analyse how ω affects the relationship between t_{bright} and e (equation 5). As expected, t_{bright} is minimum for the toy model ($\omega = 0^\circ$) and reaches a maximum value

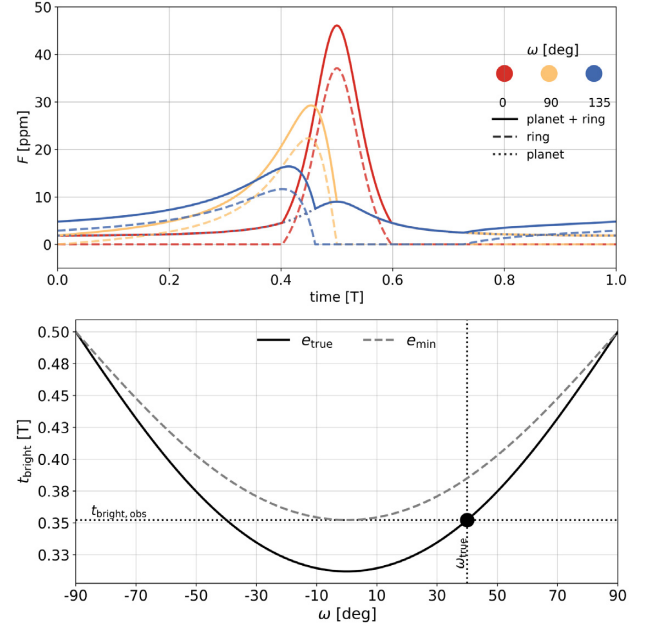


Figure 5. Upper panel: Normalized flux as a function of time. The contributions of the planet (dotted lines), ring (dashed lines), and planet + ring (solid lines) are shown. The colours represent different values of the orbital periastron argument (ω) in Fig. 1. Lower panel: Relationship between the bright time (t_{bright}) and the eccentricity (e) given by equation (5).

when the summer solstice occurs at quadrature ($\omega = 90^\circ, -90^\circ$). Since an optimal orbital configuration was assumed in the toy model, equation (8) and the observed t_{bright} will underestimate the true eccentricity of the orbit (see the dashed curve in the lower panel of Fig. 5). However, even in a more realistic situation, equation (8) is still valid as a formula for constraining e :

$$e \geq e_{\text{min}} = \frac{\pi}{2} \left(\frac{1}{2} - \frac{t_{\text{bright}}}{T} \right). \quad (14)$$

Also, an unknown orbital inclination contributes to either increasing or reducing F_{dark} and F_{max} with respect to their corresponding values in the toy model (equations 9 and 11, respectively). In the case of F_{dark} , ω tends to increase its true value, while i may either reduce it or increase it, thereby compensating the effect of ω . On the other hand, the constraint given by equation (10) will not be substantially modified (at least within a first-order estimation of A_{planet}). By comparison with the toy model, both ω and i lower the measured value of F_{max} . This implies that equation (13) would remain valid, although the minimum A_{ring} will probably be underestimated.

To find *fiducial* parameters of real systems, a more general and complex model for fitting stellar light curves needs to be developed. To that end, we should include second-order effects such as the shadow/shine of the planet's surface and atmosphere, forward scattering, the opacity of the rings, the polarization and reflection of light by dusty/icy ring particles, etc. For this *optical complexity*, we propose (and intend) to create a scattering model centred on the ring (most models are centred on the star), where the properties of the scattered light are computed in the sky above the planetary rings after describing the astronomical position of the star, the planet, and the observer. However, such a model is the subject of an ongoing research that will be introduced in a forthcoming paper.

5 SUMMARY AND CONCLUSIONS

The first-order semi-analytical model presented here demonstrates that rings may produce detectable and distinguishable photometric signatures on the light curve of non-transiting exoplanets. We have quantified the phase-dependent amount of light that a tilted ringed planet can reflect towards a distant observer, and the model is described through four main morphological features of the so-called *bright curve* that help us derive some analytical constraints to orbital and physical parameters of the system. This model can be useful to detect (ringed) planets with large ring surfaces lying in face-on orbits, whose survival time-scale is ~ 1 Myr (Schlichting & Chang 2011). Thus, our method is complementary to transit photometry, which is rather biased in favour of edge-on configurations.

It is worth noting that we considered a system composed of a Saturn-like planet in a close-in orbit around a Sun-like star. This is a simplified scenario for the sake of presenting our model, but under this assumption, transits are still likely to occur: for an orbit of 0.1 au, a Saturn-like planet would have a transit probability of ~ 10 per cent. However, the remaining 90 per cent of the exoplanets of this specific population cannot be detected via transit photometry, so the model proposed here would be ideal for complementing the general distribution of close-in exoplanets.

The applicability of our model would be further ensured if we had independent *astrometric* measurements of stellar masses (M_*). However, astrometry is still a developing area with 5 candidates awaiting confirmation and only 10 confirmed planets. By following up on those systems and using the inferential procedure presented in Section 3, one might reveal the presence of ringed planets and help constrain some of their characteristics. Moreover, the next releases of data from *Gaia* will increase the number of discoveries via astrometry and thus the chance of finding and characterizing the yet unseen population of exorings.

Within the scope of this work, and after some assumptions on the radius of the central body and the shape/orientation of the rings, we simulated the system 1SWASP J140747 (Kenworthy & Mamajek 2015). Regardless of its low stellar luminosity and large planetary semimajor axis ($a_p \sim 3.9$ au), we found that the maximum amount of scattered light from the system rounds the 150 ppm level at the orbital periastron ($e \sim 0.6$), becoming detectable with current instrumentation. Also, in analogy with HD 189733b (Knutson et al. 2007), a modulation between consecutive stellar flux peaks should be expected. This is produced due to the scattered light from the ring, which induces a phase-modulated shift in the stellar light curve.

The signal produced by scattered light from face-on close-in planets may be meagre and misleading when the signal-to-noise ratio (SNR) of observations is considerably low. This, however, can be circumvented by adopting state-of-the-art methods to increase the SNR (e.g. Colón et al. 2012; Stefansson et al. 2017). With high instrumental sensitivity levels, phase-dependent deviations of the scattered starlight only from the planet (see the dotted lines in the lower panel of Fig. 5) can help constrain different orbital parameters using analytical expressions of the variable flux, $F(t)$.

A striking result of the phenomena described here is that a bright curve resulting from the scattered light from planetary rings is an indirect way of characterizing the planet's interior/atmosphere, which is apposite in the *TESS* (present) and *JWST* (future) observational era. Because of its broad wavelength coverage and high sensitivity, the *JWST* will be a powerful tool for exoplanet transit photometry that should allow us to achieve the required precision of 10–100 ppm for either transit or flux peak detection.

Finally, if we measure the polarized light of aligned dust grains in the rings, embedded into an undetected exoplanet's magnetic field,

we can also constrain the planet's rotation and composition. The evidence of those magnetic fields may help verify and improve current models of magnetic field formation and evolution. To summarize, the still unknown population of exorings, in particular, those discovered by scattered light, might be a proxy whereby we can further characterize and constrain the orbital and physical properties of close-in giant exoplanets.

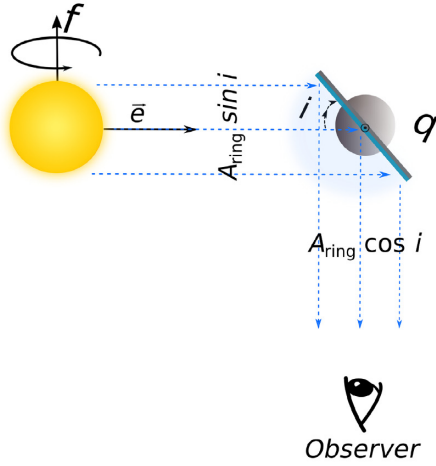
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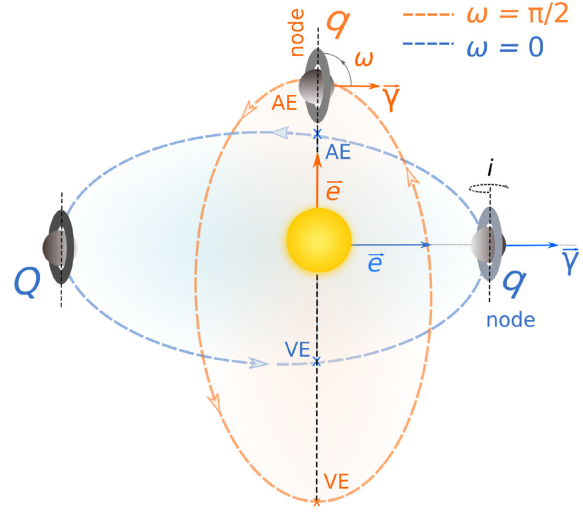
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(a) Perpendicular to the observer's view



(b) Observer face-on view

**Figure A1.** (a) Schematic illustration of the system considered in this work. (b) Schematic of two different orbital orientations on the sky plane.

APPENDIX A: SYSTEM GEOMETRY

For study purposes, the system considered in this letter is composed of a nearby ringed planet whose orbital plane is coplanar to that of the sky. The ring system around the planet has a constant inclination i , as shown in Figs 1 and A1, where $i = 0$ means that the rings' and orbital planes are coplanar. Also, the system studied in Section 4 assumes an orbital orientation $\omega = 0^\circ$ where the Laplace–Runge–Lenz vector $\hat{e}$, whose magnitude is the orbital eccentricity, is aligned with γ , which is a vector perpendicular to the line of the nodes produced by the intersection between the orbital and rings' planes (see the blue orbit in the left-hand panel of Fig. A1). Also, an orange orbit is depicted in the right-hand panel of Fig. A1, but in this case $\omega = 90^\circ$, meaning that $\gamma \perp e$.

An equivalent approach consists in rotating the ring around a vector normal to the plane that passes through its centre. In any case, ω pinpoints the changes of the ring's orientation leaving the inclination i constant. However, we choose the former definition (i.e. that of the previous paragraph) since it allows us to derive/constrain other orbital parameters such as the orbital eccentricity by using information that comes out from the reflected light by the ring.

Regarding the above, $\omega = 0^\circ$ (where the planetary true anomaly $f = 0$) makes the rings to appear brighter at the periastron as compared to other configurations, since the area of the rings projected towards the star is maximum ($A_{\text{ring}} \sin i$). In other points of the orbit (i.e. $f \neq 0$), such a projected area is modulated by Φ , defined in equation (3), which decreases during the equinoxes where $\hat{n}_s \cdot \hat{r} = \cos f = \cos \pm \pi/2 = 0$. Thus, the time-dependent projected area of the rings towards the star is defined as $A_{\text{ring}}(t) = A \sin i \Phi(t)$. It is important to note that the line connecting the equinoxes passing through the star (see the black dashed line in Fig. 1) is parallel to the line of the nodes defined previously.

To conclude, since the system is assumed to be face-on aligned, only a portion of the reflected photons reaches the observer. To account for this effect, we need to multiply the aforementioned projected area by $\cos i$, so we obtain that the intensity of the observed light, presented in equation (4), must be given by

$$I_{\text{ring}}(t) = I_\star A_{\text{ring}}(t) B_\star(t) = \frac{I_\star}{4\pi r(t)^2} A \sin i \cos i \Phi(t). \quad (\text{A1})$$

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